

Chapter 1 — Abstract setting

Variational form, well-posedness and Sobolev
spaces

MTH3340 · Numerical Methods for PDEs

1 Abstract variational form

Definition (Variational problem)

A variational problem reads as $u \in V : a(u, v) = \ell(v), \quad \forall v \in V_0$, where V_0 is a vector space of functions, $V = u_0 + V_0$ is an affine space of functions, $a : V \times V_0 \rightarrow \mathbb{R}$ is a bilinear form and $\ell : V_0 \rightarrow \mathbb{R}$ is a linear form.

V : **trial space** (where we seek the solution); V_0 : **test space** (v.s. of functions we use to “test” the equation).

As above, it can be re-stated using the offset function method, to have a trial space that is a vector space,

$$u = u_0 + \tilde{u}, \quad \tilde{u} \in V_0 : a(\tilde{u} + u_0, v) = \ell(v) \quad \forall v \in V_0,$$

i.e., $a(\tilde{u}, v) = \tilde{\ell}(v) \doteq \ell(v) - a(u_0, v)$.

This problem is obtained from the QMP using calculus of variations (previous lecture).

2 Functional analysis

Idea: find the *largest* space for which the bilinear form is bounded (“makes sense”) & satisfies suitable bc’s (for uniqueness).

A minimiser must have bounded energy.

Definition (Maximal functional space)

$$V \doteq \{v : \Omega \rightarrow \mathbb{R} : a(v, v) < \infty\}$$

is the **maximal functional space** wrt a (the largest space for which J makes sense).

3 The $L^2(\Omega)$ space

For the mass bilinear form, i.e., $J_0(u) \doteq \int_{\Omega} u^2$, the maximal functional space is

$$V_0 = \left\{ v : \Omega \rightarrow \mathbb{R} : \int_{\Omega} v^2 < \infty \right\}.$$

Definition ($L^2(\Omega)$ space)

$L^2(\Omega)$ is the space of *square-integrable* functions on Ω . It is a normed space for

$$\|v\|_{L^2(\Omega)} \doteq \left(\int_{\Omega} |v(x)|^2 dx \right)^{1/2}.$$

Ex.: bc's are ill-posed in $L^2(\Omega)$ (or analogously, pointwise evaluations, e.g., $u(x)$, $x \in \Omega$).

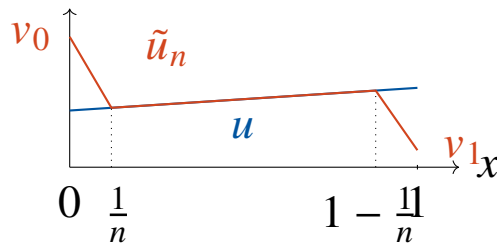
Bc's cannot be imposed in $L^2(\Omega)$

Example (Boundary conditions cannot be imposed in $L^2(\Omega)$)

Take $u \in C^0([0, 1]) \subset L^2((0, 1))$ and arbitrary values v_0, v_1 , and build the sequence of perturbations

$$\tilde{u}_n \doteq \begin{cases} u(x) + (1 - nx)(v_0 - u(0)), & 0 \leq x \leq 1/n, \\ u(x), & 1/n < x < 1 - 1/n, \\ u(x) - n(1 - 1/n - x)(v_1 - u(1)), & 1 - 1/n < x \leq 1. \end{cases}$$

Clearly $\tilde{u}_n(0) = v_0$ and $\tilde{u}_n(1) = v_1$ for any n , but $\|\tilde{u}_n - u\|_{L^2((0,1))}^2 = \frac{1}{3n}((v_0 - u(0))^2 + (v_1 - u(1))^2) \rightarrow 0$ as $n \rightarrow \infty$ (check it!).



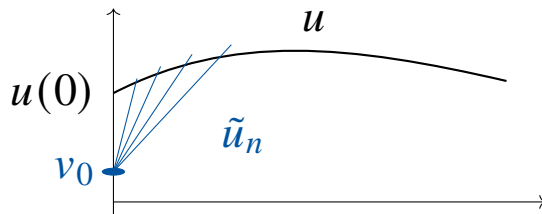
The L^2 -norm is not strong enough to *feel* the bc's.

The space with bc's is not closed!
 $\lim_{n \rightarrow \infty} \tilde{u}_n = u \notin \text{the space.}$

Pointwise evaluations are not continuous in L^2

Another way to state this fact: pointwise evaluations are not continuous in L^2 . E.g., in the example above, $\|\tilde{u}_n - u\|_{L^2} \rightarrow 0$ but

$$|\tilde{u}_n(0) - u(0)| = |v_0 - u(0)| \geq C > 0.$$



$$\implies |\tilde{u}_n(0) - u(0)| \leq c \|\tilde{u}_n - u\|_{L^2(\Omega)}$$

is false!

(Continuity of a map: $|f(u) - f(v)| \leq c |u - v|$.)

Cauchy sequences and completeness

Definition (Cauchy sequence)

Consider a normed vector space V equipped with the norm $\|\cdot\|$. A sequence $\{v_n\}_{n \in \mathbb{N}}$ of elements of V is called a Cauchy sequence if

$$\forall \epsilon > 0 \exists n = n(\epsilon) \in \mathbb{N} : \|v_k - v_m\| \leq \epsilon, \quad \forall k, m \geq n.$$

Note: obviously, convergent seq. $\longrightarrow$ Cauchy seq.: if $\lim_{n \rightarrow \infty} u_n = u \in V$,

$$\lim_{n, m \rightarrow \infty} \|u_n - u_m\| \leq \lim_{n \rightarrow \infty} \|u_n - u\| + \lim_{m \rightarrow \infty} \|u_m - u\| = 0.$$

Ex.: $\mathbb{R}$ with $|\cdot|$ (abs. value) is complete (axiom); finite-dim spaces are complete; $C^0(\bar{\Omega})$ wrt the $\|\cdot\|_\infty$ norm is complete ($\|u\|_\infty = \sup_{x \in \bar{\Omega}} |u(x)|$); $C^0(\bar{\Omega})$ wrt the $\|\cdot\|_{L^2}$ norm (**too weak**) is *not* complete (the ramps of the previous lecture!).

Completeness is a property of the normed space (space of functions AND norm being used).

Banach and Hilbert spaces

COMPLETENESS is the essential ingredient for $\exists!$ of the VP. We define spaces with this property:

Definition (Banach space)

A *normed* vector space is called *complete* if every Cauchy sequence converges. A complete normed vector space is called a *Banach space*.

Definition (Hilbert space)

If a Banach space is endowed with a norm that is an energy norm with respect to a symmetric positive definite bilinear form (an *inner product*), it is called a *Hilbert space*.

Ex.: $L^2(\Omega)$ is *complete* wrt $\|v\|_{L^2(\Omega)} \doteq \left(\int_{\Omega} v^2\right)^{1/2}$ (stated without proof):
a Hilbert space with inner product $(u, v)_{L^2(\Omega)} \doteq \int_{\Omega} uv$.

Well-posedness in Hilbert spaces

With these ingredients, we can prove $\exists$!

Theorem (Existence and uniqueness of solutions in Hilbert spaces)

Let us consider a Hilbert space V_0 endowed with the inner product $a : V_0 \times V_0 \rightarrow \mathbb{R}$ and a continuous linear functional $\ell : V_0 \rightarrow \mathbb{R}$. The corresponding quadratic minimisation problem $u = \operatorname{argmin}_{v \in V_0} J(v)$, $J(v) \doteq \frac{1}{2}a(v, v) - \ell(v)$, has a unique solution.

The proof is accessible, see below.

Proof of existence

Proof. J is bounded below (previous lecture), so we can define a sequence $\{v_n\}_{n \in \mathbb{N}}$ such that $|J(v_n) - \mu| \leq 1/n$, $\mu \doteq \inf_{v \in V_0} J(v)$. Due to the bilinearity of a ,

$$\frac{1}{2}(J(v) + J(w)) - J\left(\frac{1}{2}(v + w)\right) = \frac{1}{8}\|v - w\|_a^2.$$

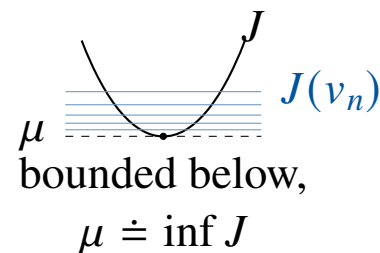
Clearly, $J\left(\frac{1}{2}(v_k + v_m)\right) \geq \mu$ (since μ is the inf), which leads to

$$\frac{1}{8}\|v_k - v_m\|_a^2 \leq \frac{1}{2}(J(v_k) + J(v_m)) - \mu \leq \max\{1/k, 1/m\}.$$

Thus, $\{v_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence and $u \doteq \lim_{n \rightarrow \infty} v_n \in V_0$ due to completeness of V_0 . Since J is a continuous functional in V_0 ,

$$J(u) = J\left(\lim_{n \rightarrow \infty} v_n\right) \stackrel{\text{by cont.}}{=} \lim_{n \rightarrow \infty} J(v_n) = \mu.$$

Thus, $u \in V_0$ is a global minimiser; uniqueness by the uniqueness lemma (previous lecture). $\square$



Note: $\inf_{x \in (0,1]} x^2 = 0$ even though $x^2 > 0$ in $(0, 1]$, i.e., the inf is not attained in $(0, 1]$.

Key part: thanks to completeness, the limit is in V_0 .

From C^0 to the L^p spaces

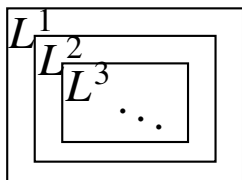
Note: the step-target example (previous lecture) was ill-posed in $C^0([-1, 1])$ but it is WP in $L^2((-1, 1))$, with minimiser g . C^0 was too small: it did not include limit points of Cauchy seq's, i.e., it was not complete.

We can also define

$$\|u\|_{L^p(\Omega)} \doteq \left[\int_{\Omega} |u(x)|^p \right]^{1/p}, \quad p \geq 1;$$

$p = 2$ is the $L^2(\Omega)$ space; one can check this is a norm for $p \in [1, +\infty)$.

On a bounded Ω : $\dots L^3(\Omega) \subset L^2(\Omega) \subset L^1(\Omega)$, since



$$\|u\|_{L^1} = \int_{\Omega} |u| \cdot 1 \leq \|u\|_{L^2} \left(\int_{\Omega} 1 \right)^{1/2} = C \|u\|_{L^2}$$

(Cauchy–Schwarz). Thus, $u \in L^2(\Omega) \longrightarrow u \in L^1(\Omega), \dots$

4 The $H^1(\Omega)$ space

Definition (Gradient of a function)

Given a function $f : \Omega \subset \mathbb{R}^d \rightarrow \mathbb{R}$, we define its gradient as $\nabla f(x) \doteq [\partial f / \partial x_1(x), \dots, \partial f / \partial x_d(x)]^T \in \mathbb{R}^d$ for $x \in \Omega$.

$$\nabla f = \begin{pmatrix} \partial f / \partial x \\ \partial f / \partial y \end{pmatrix} \text{ in 2D}$$

Definition (Sobolev space)

A Sobolev space is a vector space of functions endowed with a norm that combines L^p norms of the function itself and its derivatives up to a given order.

E.g., u such that $u \in L^2(\Omega)$, $\nabla u \in [L^2(\Omega)]^d$
 $\longrightarrow u \in H^1(\Omega)$.

$$\boxed{L^2 \supset H^1}$$

The spaces $H_0^1(\Omega)$ and $H^1(\Omega)$

Definition (Sobolev space $H_0^1(\Omega)$)

The space of square integrable functions with square integrable gradients on Ω that *vanish on* $\partial\Omega$ is denoted by $H_0^1(\Omega)$. It is a normed space for $|v|_{H^1(\Omega)} \doteq (\int_{\Omega} |\nabla v(x)|^2 dx)^{1/2}$.

Not a norm without bc's (the norm of a constant would be 0!) H^1
“feels” bc's: $H_0^1(\Omega)$ makes sense (unlike “ L_0^2 ”).

Definition (The $H^1(\Omega)$ space)

The Sobolev space $H^1(\Omega) \doteq \{v \in L^2(\Omega) : \int_{\Omega} |\nabla v(x)|^2 dx < \infty\}$ is a normed space with $\|v\|_{H^1(\Omega)}^2 \doteq \|v\|_{L^2(\Omega)}^2 + |v|_{H^1(\Omega)}^2$.

Adding the L^2 norm, no need for bc's.

The H^1 spaces are Hilbert spaces

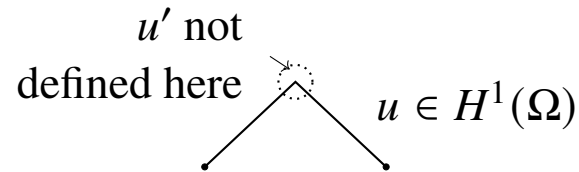
$H^1(\Omega)$ and $H_0^1(\Omega)$ are *complete* (stated without proof): Hilbert spaces with inner products

$$(u, v)_{H^1(\Omega)} \doteq \int_{\Omega} uv + \nabla u \cdot \nabla v \quad \text{and} \quad (u, v)_{H_0^1(\Omega)} \doteq \int_{\Omega} \nabla u \cdot \nabla v.$$

That is all we need for the model problem of Lecture 1: its bilinear form $\int_{\Omega} u'v'$ is the inner product of $H_0^1(\Omega)$, so the well-posedness theorem gives $\exists!$ (continuity of ℓ : next lecture!).

$H^1(\Omega)$ functions need not be smooth

$$H^1(\Omega) \not\subset C^1(\Omega)$$



Example (Piecewise continuous functions in $H^1(\Omega)$)

$H^1(\Omega)$ includes functions that do not possess classical derivatives, i.e., they are not differentiable at all points. For instance, consider the piecewise function in $[0, 1]$

$$u(x) = \begin{cases} 2x, & 0 < x < 1/2, \\ 2(1-x), & 1/2 \leq x < 1. \end{cases}$$

This function belongs to $H^1(\Omega)$ since $|u|_{H^1(\Omega)}^2 = \int_0^1 |\nabla u(x)|^2 dx = 4 < \infty$. In general, the space $C_{pw}^1(\bar{\Omega})$ of continuous functions with piecewise continuous first derivatives is a subset of $H^1(\Omega)$.

VERY IMPORTANT for FEM. We use these functions for approximation.